

LISTENING TO UAV: 3D TRAJECTORY ESTIMATION VIA ACOUSTIC TRANSFORMER

Luyang Yin^{1,*}, Yanzhang Feng^{2,*}, Yu Zhao^{2,†}, Haoyu Wang⁵, Hongzhi Wang², Fan Yang², Hao Peng³, Huinan Yang^{4,5}

¹Shandong University of Science and Technology, Jinan, China

²Zhejiang University, Hangzhou, China

³University of Complutense, Madrid, Spain

⁴University of Shanghai for Science and Technology, Shanghai, China

⁵Tongji University, Shanghai, China

ABSTRACT

3D trajectory estimation of moving sound sources is critically important for applications in UAV surveillance, robotic environmental perception, and immersive audio systems. This task remains challenging due to complex acoustic environments characterized by noise, reverberation, and source mobility. Traditional multi-stage methods suffer from error accumulation and require extensive parameter tuning, while existing deep learning approaches often use suboptimal acoustic representations that discard essential physical information. To address these limitations, we propose an end-to-end framework that integrates physical principles into the learning process. Our method employs complex spectra and GCC-PHAT features to preserve crucial time-delay information, and introduces a hybrid CNN-Transformer encoder to extract both local and global acoustic features. A novel cross-attention mechanism enables differentiable physical reasoning between acoustic observations and trajectory states. Extensive evaluations on the MMAUD dataset demonstrate that our approach significantly outperforms existing methods in estimation accuracy and robustness across diverse acoustic conditions.

Index Terms— UAV trajectory estimation, anti-UAV, sound source localization, acoustic signal processing

1. INTRODUCTION

Sound source localization and **3D trajectory estimation** technologies play a crucial role in anti-UAV research and applications, including drone surveillance [1], intelligent robotic perception, and virtual reality systems. The accurate detection and tracking of unauthorized UAVs present significant challenges in complex acoustic environments characterized by noise interference, reverberation effects, and dynamic source movements [2, 3].

Conventional sound source localization methods typically follow multi-stage processing pipelines that include

time delay estimation, geometric localization, and trajectory filtering components [4, 5]. While these approaches offer physical interpretability, they suffer from two fundamental limitations: error accumulation through cascaded processing modules, and heavy reliance on manual parameter tuning, which severely restricts their practical applicability in real-world anti-UAV scenarios [6].

Recent advances in deep learning have introduced end-to-end learning paradigms for sound source localization [7, 8]. However, current deep learning approaches exhibit notable shortcomings. Most methods employ **Mel-spectrograms** as input features [9, 10], which discard essential phase information and distort critical time-delay cues through nonlinear frequency compression, fundamentally limiting localization accuracy [11]. Architecturally, convolutional neural networks struggle to capture long-term temporal dependencies [12], recurrent networks demonstrate inefficiency in modeling global time-frequency relationships [13], and pure Transformer architectures require large-scale pre-training while lacking appropriate inductive biases for acoustic processing [14].

To address these challenges, we propose an end-to-end sound source trajectory estimation framework that incorporates physical priors into the learning process. Our approach introduces three principal innovations:

1. We develop a physically-informed input representation utilizing **complex spectra** and **GCC-PHAT** features [4] that explicitly preserve critical time-delay information essential for accurate localization.
2. We design a **CNN-Transformer hybrid** encoder architecture that effectively extracts local acoustic features while modeling global contextual relationships, enabling comprehensive spatio-temporal feature learning.
3. The complete system implements **end-to-end optimization** through differentiable cross-attention mechanisms that integrate acoustic evidence with trajectory kinematics, avoiding error accumulation while maintaining physical interpretability.

*These authors contributed equally to this work.

†Corresponding Author: Yu Zhao, email: y.zhao@zju.edu.cn

2. METHOD

2.1. Data Preprocessing and Physical Feature Extraction

Objective: To transform raw multi-channel audio signals into an information-rich tensor representation that explicitly encodes both energetic and phase-difference spatial cues essential for sound source localization.

The input to our system consists of four-channel synchronous time-domain audio signals. The preprocessing pipeline begins with segmenting the continuous audio stream into short Hanning windows of 400 ms duration with 75% overlap [5]. For each channel within every window, we compute the complex-valued spectrogram via the Short-Time Fourier Transform (STFT), obtaining both magnitude and phase information across time and frequency:

$$S_c(f, t) = |S_c(f, t)| \cdot e^{j\phi_c(f, t)} \quad \text{for } c = 1, 2, 3, 4. \quad (1)$$

From these complex spectrograms, we construct a six-channel feature tensor $\mathcal{X}(t) \in \mathbb{R}^{6 \times F \times T}$ that serves as the input to our neural network. The first four channels contain the *log-magnitude spectra* for each microphone channel, denoted as $\mathcal{X}_{\log}(t) = \log(|S_c(f, t)| + \epsilon)$. The remaining two channels contain *Generalized Cross-Correlation with Phase Transform (GCC-PHAT) spectra* [4], denoted as $\mathcal{X}_{\text{GCC-PHAT}}(t)$, computed for two carefully selected microphone pairs. The GCC-PHAT representation provides enhanced time-difference-of-arrival (TDOA) information that is robust to environmental acoustics and particularly valuable for spatial localization.

2.2. Encoder Architecture: Hierarchical Acoustic Feature Extraction

Objective: To extract and fuse multi-scale acoustic features from the input tensor, transforming local spectral patterns into a global contextual representation suitable for cross-attention mechanisms.

The encoder network $\mathcal{E}_\theta$ implements a hierarchical feature extraction process that maps the input tensor to a sequence of contextualized embeddings:

$$\mathcal{E}_\theta : \mathbb{R}^{6 \times F \times T} \rightarrow \mathbb{R}^{L \times D_{\text{model}}}, \quad (2)$$

where $L = F' \times T'$ represents the sequence length derived from reduced spatial dimensions, and D_{model} denotes the hidden dimension size.

The encoding process begins with a CNN backbone architecture based on ResNet principles [12]. This network processes the input feature tensor as an acoustic image, operating on the frequency-time plane through successive convolutional layers. The CNN extracts increasingly abstract features—from basic spectral edges and textures in early layers

to complex patterns such as Doppler shift trajectories and harmonic structures in deeper layers. Through strided convolutions and pooling operations, the spatial dimensions are progressively reduced from (F, T) to (F', T') , while the channel dimension increases, outputting a condensed feature map $\mathbf{F}_{\text{CNN}} \in \mathbb{R}^{C \times F' \times T'}$.

The feature map is then flattened along the spatial dimensions to form a sequence of tokens $\mathbf{Z}_0 \in \mathbb{R}^{L \times C}$, where $L = F' \times T'$. To preserve the spatial structure of the original feature map, we incorporate learnable positional embeddings. We define separate frequency and time positional embeddings $\mathbf{E}_f \in \mathbb{R}^{F' \times D_{\text{emb}}}$ and $\mathbf{E}_t \in \mathbb{R}^{T' \times D_{\text{emb}}}$, which are expanded and combined through broadcasting to match the spatial dimensions and then added to the flattened sequence:

$$\mathbf{Z} = \mathbf{Z}_0 + \text{Broadcast}(\mathbf{E}_f) + \text{Broadcast}(\mathbf{E}_t). \quad (3)$$

This position-augmented sequence is processed by a standard Transformer encoder [13] comprising multiple layers of multi-head self-attention and position-wise feed-forward networks. The self-attention mechanism enables each token to attend to all other tokens in the sequence, capturing global dependencies across the entire acoustic context. The output of this process is the final acoustic context representation $\mathbf{H}_{\text{enc}} \in \mathbb{R}^{L \times D_{\text{model}}}$, which serves as the source for Keys and Values in the decoder's cross-attention mechanism.

2.3. Decoder Architecture: Spatio-Temporal Reasoning

Objective: To estimate the current 3D sound source position by integrating historical trajectory information with acoustic evidence through an autoregressive decoding process.

The decoder network $\mathcal{D}_\phi$ implements an autoregressive mapping from historical states and acoustic context to current position:

$$\mathcal{D}_\phi : \{\mathbf{s}_{t-K}, \dots, \mathbf{s}_{t-1}\} \times \mathbf{H}_{\text{enc}} \rightarrow \mathbb{R}^3, \quad (4)$$

where K represents the historical trajectory length, and $\mathbf{H}_{\text{enc}}$ is the encoded acoustic representation from the encoder.

The decoder comprises a stack of identical layers, each containing three specialized sub-layers. The input historical trajectory coordinates $\{\mathbf{s}_{t-K}, \dots, \mathbf{s}_{t-1}\}$ are first embedded into a continuous space of dimension D_{model} and augmented with sinusoidal positional encodings to preserve temporal ordering, resulting in $\mathbf{S}_{\text{emb}} \in \mathbb{R}^{K \times D_{\text{model}}}$.

The first sub-layer employs masked multi-head self-attention, where each position in the historical sequence can only attend to previous positions. This masking ensures causality in the autoregressive process and enables the model to learn kinematic priors and motion patterns from the trajectory history.

The second sub-layer performs multi-head cross-attention, which forms the core of our physical reasoning mechanism. The output from the self-attention sub-layer, $\mathbf{H}_{\text{self}} \in$

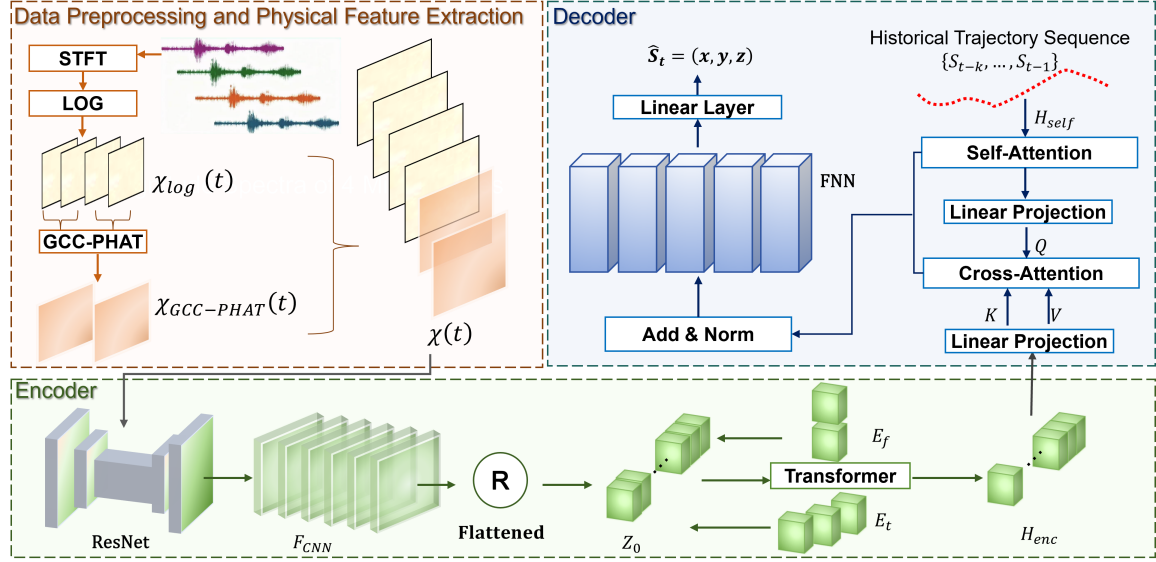


Fig. 1. Pipeline. The proposed neural acoustic field framework processes four-channel microphone array inputs through a dedicated preprocessing stage that computes both log-magnitude and GCC-PHAT spectral features to form a 6-channel feature tensor. This tensor is encoded via a hierarchical CNN-Transformer architecture that extracts and fuses local and global spatio-temporal features. The decoder subsequently performs trajectory estimation by integrating encoded acoustic representations with historical states through a cross-attention mechanism, enabling direct regression of 3D sound source coordinates.

$\mathbb{R}^{K \times D_{\text{model}}}$, serves as the Query. The acoustic context representation $\mathbf{H}_{\text{enc}}$ is linearly projected to form the Key and Value matrices:

$$\mathbf{Q} = \mathbf{H}_{\text{self}} \mathbf{W}^Q, \quad \mathbf{K} = \mathbf{H}_{\text{enc}} \mathbf{W}^K, \quad \mathbf{V} = \mathbf{H}_{\text{enc}} \mathbf{W}^V, \quad (5)$$

where $\mathbf{W}^Q, \mathbf{W}^K, \mathbf{W}^V \in \mathbb{R}^{D_{\text{model}} \times D_k}$ are learnable projection matrices. The cross-attention is computed as:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax} \left(\frac{\mathbf{Q} \mathbf{K}^T}{\sqrt{d_k}} \right) \mathbf{V}. \quad (6)$$

This mechanism enables the decoder to dynamically query the acoustic evidence most relevant for each historical state when estimating the current position.

The third sub-layer consists of a position-wise feed-forward network, which is a two-layer multilayer perceptron with ReLU activation. This network applies an identical nonlinear transformation to each position in the sequence. Residual connections and layer normalization surround each sub-layer to facilitate stable training of deep networks.

The output of the final decoder layer is projected through a linear layer to produce the estimated 3D coordinates $\hat{\mathbf{S}}_t \in \mathbb{R}^3$. The entire network is trained end-to-end using Smooth L1 loss between predicted and ground-truth trajectories, enabling joint optimization of all parameters through backpropagation.

3. EXPERIMENT

3.1. Experimental Setup

Dataset. We use the MMAUD dataset [15], which contains synchronized 41.8 kHz four-channel audio and millimeter-accurate 3D trajectory annotations. With realistic environmental noise, it is split non-chronologically into a 7:3 train/test ratio to avoid data leakage and ensure thorough evaluation.

Evaluation Metrics. Metrics include axis-wise absolute errors (D_x, D_y, D_z) for L1 distances per coordinate, and absolute trajectory error (APE) for the average Euclidean distance across full trajectories.

Baseline Methods. We compare with state-of-the-art methods from three modalities: visual-based methods (VisualNet [16], DarkNet [17]) that suffer in dark conditions; audio-based methods including AudioNet [16], DroneChase [20], and TAME [21]; and audio-visual fusion models TalkNet [18], AV-PED [16], and AV-FDTI [19].

Implementation Details. Training is performed on an RTX 3090 GPU using the Adam optimizer (initial learning rate 1×10^{-4} , batch size 32) for 100 epochs.

3.2. Comparison to State-of-the-Art

Our method outperforms all existing approaches. Quantitative and qualitative results are presented in Table 1 and Figure 2, respectively.

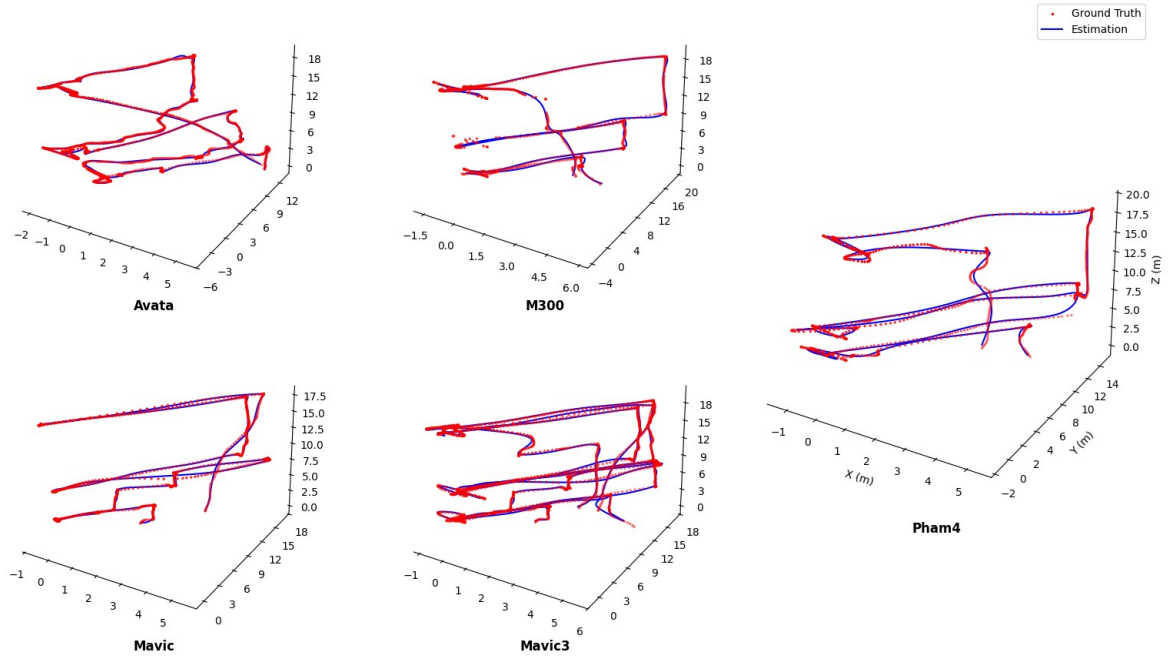


Fig. 2. Comparison between predicted trajectories and GT trajectories on the MMAUD dataset.

Table 1. 3D trajectory estimation and accuracy on the MMAUD V1 Dataset [15]. Best results are highlighted in **bold**, and second best are underlined.

Modality	Network	Light				Dark				$\overline{APE}$
		D_x	D_y	D_z	APE	D_x	D_y	D_z	APE	
Visual	VisualNet [16]	0.24	0.39	0.32	0.65	1.98	6.10	8.13	11.45	6.05
	DarkNet [17]	0.23	0.46	<u>0.23</u>	0.63	1.84	5.50	4.57	8.31	4.47
Audio-Visual	TalkNet [18]	0.31	0.69	0.44	0.99	1.13	3.39	3.92	5.82	3.41
	AV-PED [16]	0.31	0.50	0.59	0.97	0.58	1.54	2.26	3.13	2.01
	AV-FDTI [19]	<u>0.13</u>	0.31	0.38	0.58	0.35	1.06	1.10	1.89	1.24
Audio	AudioNet [16]	0.60	1.76	1.59	2.80	0.60	1.76	1.59	2.80	2.80
	DroneChase [20]	0.54	1.59	1.51	2.64	0.54	1.59	1.51	2.64	2.64
	TAME [21]	0.11	<u>0.30</u>	0.34	<u>0.55</u>	0.11	<u>0.30</u>	<u>0.34</u>	<u>0.55</u>	<u>0.55</u>
	Ours	0.15	0.23	0.21	0.35	<u>0.15</u>	0.23	0.21	0.35	0.35

Table 2. Ablation studies on different components of our proposed method. APE (m) results are reported on the MMAUD test set. Best results are highlighted in **bold**.

Component	Variant	$\overline{APE}$
Input Features	Mel-Spectrogram	0.78
	LogMag + GCC (Ours)	0.35
Positional Embedding	Without PE	0.68
	With PE (Ours)	0.35
Encoder Architecture	CNN-only	0.62
	Transformer-only	0.59
	CNN-Transformer (Ours)	0.35

3.3. Ablation Studies

Ablations validate our key designs (Table 2): (1) Log-Mag+GCC outperforms Mel-spec via phase preservation; (2) TF positional embeddings are critical for spatial reasoning; (3) CNN-Transformer hybrid outperforms CNN-only (ResNet-18) and Transformer-only architectures.

4. CONCLUSION

This paper introduces a novel neural acoustic field framework for 3D trajectory estimation, achieving state-of-the-art performance on the MMAUD dataset and enabling robust acoustic perception in complex environments.

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